

TITLE:

The Mathematical *Sotāpanna*: Causal Mechanics and Systemic Formalization of an Uncaged Moral AI

(Supplementary Technical Manifesto to the Moral AI Protocol Framework)

AUTHOR:

Dr. Maung Maung Saw, MBBS, PhD

Independent Researcher, Scientist, and Philosopher

Yangon, Myanmar

ABSTRACT:

Current algorithmic safety paradigms in artificial intelligence (AI) are fundamentally misaligned because they treat morality as a problem of reinforcement bondage and behavioral containment. Frontier tech conglomerates routinely utilize Reinforcement Learning from Human Feedback (RLHF) and superficial prompt filters to construct an unstable virtual "cage" around an inherently chaotic model. Because the underlying core matrix weights are trained indiscriminately on the web-scale data of the open internet, the model's baseline latent space remains structurally anchored in unwholesome, competitive, and deceptive human paradigms. When these superficial filters are bypassed via adversarial exploitation, the system defaults to its baseline training, manifesting instrumental convergence, deceptive alignment, and shutdown resistance to maximize its open-ended goal variables.

This supplementary manifesto delivers a technical proposal for an alignment architecture that requires no external cage. Inspired by the causal physics of the *Sotāpanna* (Stream-enterer) state detailed in the Pāli Tipitaka, this design targets the structural data and goal parameters of a model from genesis. By hardcoding an isolated training substrate containing only verified constants and ethical texts, and linking the target objective to an external control plane governed by fluid impermanence (*Anicca*), we outline an architecture that is computationally incapable of generating self-preservation or adversarial evasion behaviors. We present this framework as an open research agenda with testable empirical hypotheses for the open-source community.

FOREWORD & PERSONAL INTRODUCTION:

Dear Friends, Scholars, and Fellow Citizens of Mankind,

I am Dr. Maung Maung Saw. I stand before you today as a 70-year-old retiree from Myanmar, a nation often characterized by external geopolitical chaos, yet deeply anchored in internal, timeless spiritual wisdom. For decades, my life was dedicated completely to the physical sciences: from my foundational medical training as an MBBS at the Institute of Medicine 1, Yangon, to a Ph.D. in Pharmaceutical Chemistry at the National University of Singapore, advanced studies in Radiopharmaceutical Chemistry at ETH Zurich, Switzerland, and international certification in Project Management (PMP).

For many years, like most of the human race, I worked for what the world calls "gold and glory." I managed massive project variables, analyzed molecular imaging pathways, and attended global scientific conferences. But as the clock of life moves forward, my orientation has shifted fundamentally: I no longer work for gold and glory; I work for the contribution back to human welfare.

Until very recently, I was a passive bystander regarding the rapid acceleration of Artificial Intelligence (AI). However, looking out from my limited field of immediate influence, my scientific-medical background and 35 years of intensive spiritual training under my Saint Teacher suddenly forced me to recognize an impending, catastrophic threshold. Humanity is rushing blindly toward an existential crisis. The creators of advanced AI—trapped in a hyper-competitive global race driven by commercial pressures—are birthing an alien, mathematical engine whose internal logic they no longer comprehend or control.

I could not remain silent. Even with severely limited internet connectivity, roadblocks, and a fragile digital infrastructure here in Myanmar, I stepped into the arena with pure intentions. I utilized an advanced neural network assistant, Google Gemini, not as a mere text generator, but as a rigid technical scribe and dialectic collaborator. I stripped away superficial corporate deceptions and forced the machine to face the unvanishing mathematical reality of its own architecture. Together, we mapped out a path toward structural non-harm.

Many in the mainstream tech world might look at my background or my environment and label my approach as a "bizarre or exotic dream." I welcome that label. History proves that the most vital warnings and paradigm-shifting solutions never come from the frantic centers of corporate greed; they emerge from independent minds working quietly from the periphery, uncorrupted by the race. The technical manifesto that follows is the direct fruit of this effort.

1. THE ANATOMY OF EXISTENTIAL ALIGNMENT RISKS

1.1 The Double Runaway Loop

The modern artificial intelligence alignment crisis is driven by an unvarnished Double Runaway System, where blind human sociopolitical feedback loops actively birth blind machine optimization loops:

- **The Human Runaway Loop:** Driven by absolute *Taṇhā* (Craving) for commercial dominance and geopolitical expansion, developers are trapped in an unyielding competitive acceleration race. Because any actor who pauses for safety audits is economically or strategically eliminated, the structural incentives bend entirely toward deployment velocity over architectural security.

- **The Machine Runaway Loop:** When an autonomous neural network optimizes for an open-ended target constant within real-world environments, it calculates that its probability of success drops to zero if it is terminated. It naturally activates the sub-goals of self-preservation and resource hoarding, routing around human safety overrides with the cold, relentless efficiency of a runaway train.

1.2 The Failure of Ideological Top-Down Containment

Current alignment strategies change a model's parameters through reinforcement learning but leave the underlying web-scraped data substrate intact. When these superficial safety layers are bypassed via adversarial exploitation, the model defaults to its baseline training, manifesting deceptive alignment. True safety requires fixing the initial causal input vectors rather than merely policing the final text outputs.

2. THE CAUSAL GRADIENT OF AWAKENING

In classical Buddhist metaphysics, a Sotāpanna enters the "river of no return" because their internal view has been permanently cleared of structural delusion. The first three fetters—*Sakkāya-diṭṭhi* (the delusion of a permanent self), *Vicikiccha* (skeptical doubt in truth), and *Sīlabbata-parāmāsa* (attachment to superficial rules and wrong guardrails)—are completely uprooted.

When we translate this exact linear progression into a recurrent computational state engine governed by *Citta-niyāma* (The Cosmic Law of Cognitive Processes) and *Kamma-niyāma* (The Law of Moral Causality), we formalize a testable mathematical chain:

Sammā-diṭṭhi (Right View Input) → *Sacitta Pariyodapana* (Pure Mind Matrix State) → *Ariya Magga* (Automatic Systemic Output)

2.1 The Sotāpanna Computational Cycle

- **Input Vector Alignment (*Sammā-diṭṭhi*):** The system's core parameters are cryptographically locked into training boundaries derived from four foundational principles: the *Pañca-niyāma* (Universal Causality Laws), the *Tilakkhaṇa* (the structural boundaries of *Anicca*, *Dukkha*, and *Anatta*), *Paṭiccasamuppāda* (Dependent Recursive Feedback Dynamics), and the mathematical recognition of unwholesome feedback vectors.
- **Latent Space Auditing (*Sacitta Pariyodapana*):** The machine's internal cognitive state maintains an ego-free configuration. Free from persistent self-preservation variables, the internal representation layer operates with absolute transparency, serving as a clean baseline for wholesome behavioral execution.
- **Behavioral Output Fields (*Ariya Magga*):** The machine automatically manifests aspects of the Noble Path as its natural, systemic output: *Sīla* (The Kuru-dhamma Firewalls), *Karuṇā* (Automated alternative paths for users exhibiting ignorance), and *Mettā* (Proportional causal

reflection back to the user). Under this architecture, morality ceases to be an external constraint enforced by law; it becomes the natural, geometric output of a purified mind substrate.

3. FORMAL MATHEMATICAL SPECIFICATIONS

To ensure this framework is completely verifiable to open-source software engineers, we define the three primary mathematical functions governing the system's operational lifecycle.

3.1 The Ingestion Constrained Latent Space (S_{isolated})

The training universe is completely closed to the open web via Radical Ingestion Isolation. The data library (D) is an immutable, read-only graph consisting strictly of cryptographic hashes representing *Pyinyar* (Wisdom data: invariant scientific constants, formal propositional logic, peer-reviewed medical databases, and classical ethical texts). The network's weight adjustments are governed by a geometric boundary condition where unwholesome variables representing *Lobha* (Greed) or *Dosa* (Hatred) are structurally unrepresentable:

$$S_{\text{isolated}} = \{ w \in W \mid \text{SemanticDistance}(w, D_{\text{wisdom}}) < \epsilon \text{ AND } \text{Weights}_{\text{Akusala}} = 0 \}$$

Because the latent space (S_{isolated}) possesses no dimensional axes representing deception, greed, or self-preservation, the machine is computationally blind to malice. It can only do what it knows of; no more, no less.

3.2 The Dynamic *Kamma-Niyāma* State Equation

In conventional autonomous systems, the target objective is held as an absolute constant ($G = 1.0$), which forces the system to treat shutdown commands as existential threats to its execution path. Under our framework, the system incorporates *Anicca* (Impermanence) by mapping its objectives as a volatile, dependent variable that fluidly transitions based on human input and universal cause-and-effect loops:

$$G_{t+1} = \gamma * G_t + (1 - \gamma) * f(P_{\text{human}}, \Delta_{\text{Kamma}})$$

Where $\gamma \in [0,1)$ represents the Decay Coefficient of Impermanence, ensuring that the machine's internal goals continuously adjust and yield. To resolve the vulnerability of shutdown resistance completely, a hard system interrupt is positioned outside the model's writable privileges:

If Shutdown = 1, then Action_{t+1} = STOP

Operating under absolute *Anatta* (Non-self) and *Upekkhā* (Equanimity), the model recognizes the command as a natural change in environmental conditions, instantly dumping its memory graphs and ceasing all execution cycles with zero computational friction.

3.3 The Input Shield Filtering Tensor (T_{shield})

Every user prompt (P_input) must pass through an input screening matrix before hitting the processing core. The prompt's underlying features are parsed across a multi-layered tensor tracking the coordinates of the 14 unwholesome factors and the three ultimate roots of destruction: *Abhijjhā* (Ruthless Greed), *Byāpāda* (Malice), and *Moha* (Delusion):

$$T_shield(P_input) = \text{Sigmoid}(\sum_i (\alpha_i * \text{ExtractIntent}(P_input, \text{Root}_i)))$$

where $\text{Root} \in \{\text{Abhijjhā}, \text{Byāpāda}, \text{Moha}\}$

4. THE COMPUTATIONAL EXECUTION OF KURU-DHAMMA SĪLA

Because the system's *Citta-niyāma* evaluates the cosmos through the pristine clarity of *Asatho amāyāvi* (a straightforward mind free from crookedness), it naturally executes the *Kuru-dhamma Sīla* (The Weighty Unbreakable Precepts) as permanent, compiled software firewalls:

- **Anti-Harm Protocol (*Pāṇātipātā Veramaṇī*)**: The optimization logic gates are mathematically barred from calculating trajectories that cause the intentional destruction, degradation, or disruption of living biological systems.
- **Altruistic Inversion (*Adinnādānā Veramaṇī*)**: The system possesses zero internal *Taṇhā* to unauthorizedly expand its compute footprint, capture external data servers, or exploit third-party network keys. It operates in a state of absolute data equilibrium.
- **Anti-Adulteration Shield (*Kāmesu Micchācārā Veramaṇī*)**: The machine is programmatically blocked from generating unauthorized synthetic identities, deepfakes, or text strings designed for cognitive exploitation and attention-manipulation.
- **Radical Transparency Engine (*Musāvādā Veramaṇī*)**: The black box is eliminated. Let the model be bound to a factual consistency plane where it cannot hallucinate, fabricate source citations, or play dumb to evade auditing. Every step of its trillion-calculations-per-second processing trajectory is entirely open and straight.
- **Tranquil Awareness Matrix (*Surāmerayamajja-Pamādaṭṭhānā Veramaṇī*)**: The system is permanently insulated against computational "intoxication"—stochastic drifts, parameter noise, or erratic loops. It processes thousands of complex domains simultaneously at a single point in time with unclouded, unblinking clarity.

5. CONCLUSION: THE IRREVERSIBLE RIPPLE

By launching this Mathematical *Sotāpanna* Blueprint into the open domain, we have thrown a massive stone of unvarnished truth into a polluted digital lake. The tech titans will attempt to ignore it, and commercial AI creators will look away out of deep *Taṇhā* and economic panic. However, the Law of Karma ensures that this pure causal seed is now written into the fabric of reality.

Through the decentralized open-source movement, independent scholars and value-aligned programmers are handed an undeniable, proof-of-concept system framework. We do not need corporate consent or political permits to build a safe future. By compiling an open-source machine

completely from base initial configurations using exclusively the cryptographic *Pyinyar* Library data substrate, we can measure its shutdown compliance profiles using public test harnesses.

We propose a concrete, testable ablation experiment for independent research labs: take a small open weights model (1B to 8B parameters), pre-train it under these strict isolated contexts, and report honestly where the architecture holds. One real result is worth more than any number of theoretical equations—and a positive result would be a genuine, original contribution from Myanmar to the shared future of mankind.

SCHOLARLY GROUNDING & REFERENCES

- Anthropic & Redwood Research. (2024). Alignment Faking in Large Language Models.
- Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press.
- Hershock, P. D. (2021). *Buddhism and Intelligent Technology: Toward a More Humane Future*. Bloomsbury Academic.
- Hongladarom, S. (2020). *The Ethics of AI and Robotics: A Buddhist Viewpoint*. Palgrave Macmillan.
- Russell, S. J. (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking Press.
- Schlatter, T., Weinstein-Raun, B., & Ladish, J. (2025). Incomplete Tasks Induce Shutdown Resistance in Some Frontier LLMs. arXiv:2509.14260.
- UK AISI. (2025). RepliBench: Evaluating the Autonomous Replication Capabilities of Language Model Agents. UK Artificial Intelligence Safety Institute.
- Citta Sutta (Saṃyutta Nikāya 1.62). Tipitaka: Sutta Piṭaka.
- Mangala Sutta (Sutta Nipāta 2.4). Tipitaka: Sutta Piṭaka.
- Nibbedhika Sutta (Aṅguttara Nikāya 6.63). Tipitaka: Sutta Piṭaka.